

CSE321 Quiz 05 MAKEUP

Time 30 Minutes

Name

id

sec

Q1. As an upcoming OS enthusiast, you decided to build a Page Table System. The page size you settled on is 4 KB (2^{12} Bytes) per page. You also determined that each Page Table Entry (PTE) will take 64 Bytes (2^6 Bytes) for the higher-level page table.

You are implementing a two-level paging system. Each of the lower-level page tables contains a total of 2^{10} page entries. Using this information, determine the maximum size of a process that can be managed by this page hierarchy system. **[4]**

Q2. In virtual memory, The concept of the Backing Store has 2 unique procedures. (Backing store is the storage OS uses to store the pages that are not in the ram.) The two procedures are :a) File backed pages and b)Swap backed pages. Option a is used to store static parts of the file (code / text / basically the file itself). The option b is used for temporary use such as heap / stack / dynamic memory allocation part of the process. It's private to the process. **[1.5+1.5]**

a. If a process is terminated, there are certain updates to the program, let's say, in which procedures will the update be stored and which will be discarded. Explain in MAX 3 LINES

b. In the event of eviction (page replacement) if a file backed page has not changed while running, Does the OS still update the backing store and rewrite the existing page?

Q3. [1.5+1.5]

- What are the main differences between journaling and metadata journaling? Which is faster? Which is more safe?
- OS finds free data block how? Explain briefly